# CONDENSED MATHEMATICS MEETS TOPOLOGICAL QUANTUM FIELD THEORY: A FORMALLY VERIFIED FOUNDATION

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Abstract. We present a formally verified investigation into using condensed abelian groups (Clausen–Scholze) as a target category for topological quantum field theories, replacing topological vector spaces which fail to be abelian. Through successive rounds of AI-assisted formal verification using Lean 4, Mathlib v4.28.0 [The24], Harmonic's Aristotle system [ABD+25], Claude, and OpenAI Codex, we establish: (1) a symmetric monoidal structure on CondensedAb via sheaf localization; (2) an abstract TQFT transfer theorem; (3) an embedding functor SemiNormedGrp $\to$ CondensedAb; (4) a complete structural profile of this embedding—faithful and left exact, but not full, not conservative, and not right exact—with explicit counterexamples; (5) a hierarchy of algebraic cobordism-presentation quotients culminating in a symmetric monoidal category and a strong braided monoidal interpretation for every commutative Frobenius datum in a symmetric target; (6) the full algebraic universal equivalence between commutative Frobenius data and strong braided functors from that symmetric source; (7) finite-boundary permutation representatives and a lawful symmetric monoidal category of component-and-genus codes receiving the algebraic source by a strong braided signature functor; (8) a rank-$n$ diagonal Frobenius theory whose torus and defined connected genus words act by $n\,\mathrm{id}$, together with a finite-labelled extension whose values are label cardinalities and whose label permutations act by Frobenius isomorphisms; (9) a conditional ordinary-shadow interface which reconstructs a supplied composite strong braided theory from its generating-circle Frobenius datum; (10) a geometric prelude with locally compatible tangent orientations, a smoothly embedded boundary parametrization of the cylinder, boundary homeomorphisms, bundled smooth collar/orientation-compatibility data, and a compact Hausdorff second-countable topological seam pushout with exact control of its identifications; and (11) an abstract ribbon-category layer in which quantum trace is cyclic, the $S$-pairing is symmetric, and quantum dimension is multiplicative.

At the checkpoint documented here, the canonical implementation comprises 16,153 lines of Lean 4 across 44 source files, with **no remaining proof placeholders** and 0 custom axioms: every formal theorem reported here is machine-checked. Representation independence for finite permutation words, arbitrary-word normal-form completeness, and any equivalence between the algebraic source and the surface normal-form category remain open. The geometric layer now carries locally compatible tangent orientations, a boundary-parametrized cylinder, a topological boundary identification, bundled collar/orientation-compatibility data, and the compact Hausdorff second-countable topological pushout of two chosen-collared pieces. It does not prove a local Euclidean-with-boundary structure for that pushout, construct a compatible smooth atlas, prove smooth composition, or form a cobordism category. The ordinary-shadow construction assumes its comparison and realization functors as input; it proves no quantum geometric Langlands or quantum Fourier–Mukai equivalence. No equivalence with the geometric oriented bordism category or non-compact Chern–Simons construction is claimed.

## Contents



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## 1. Introduction

1.1. **The Homological Obstruction in Infinite-Dimensional TQFT.** A topological quantum field theory, in Atiyah's formulation [Ati88], is a symmetric monoidal functor $Z : \mathrm{Cob}_{d+1} \to C$ from a cobordism category to some symmetric monoidal target category $C$. For finite-dimensional state spaces (Dijkgraaf–Witten [DW90], Reshetikhin–Turaev [RT91], and Turaev–Viro theories), the target $C = \mathrm{Vect}$ works perfectly.

The physically interesting theories force infinite-dimensional state spaces. Chern–Simons theory [Wit88] with non-compact gauge group $\mathrm{SL}(2, \mathbb{C})$, quantum gravity partition functions, and path integrals over infinite-dimensional moduli all produce state spaces $Z(\Sigma)$ that cannot live in finite-dimensional Vect. The natural target becomes topological vector spaces.

Here a precise obstruction arises for homological and derived enhancements. The category TopAb of topological abelian groups is *not abelian*: a continuous bijective homomorphism need not have a continuous inverse, so a morphism can be simultaneously monic and epic without being an isomorphism. Consequently kernels, cokernels, exact sequences, and derived functors do not have the formal behavior available in an abelian category. This obstructs exact-sequence methods one may wish to combine with cutting and gluing. It is not an obstruction to the bare Atiyah–Segal axiom: an ordinary TQFT requires functorial composition and a symmetric monoidal product, not an abelian target.

Standard completed tensor products on topological vector spaces need not be exact: tensoring a short exact sequence with a topological vector space can destroy exactness. Disjoint union of boundaries still asks only for a monoidal product; exactness is an additional property needed when gluing calculations are organized through exact sequences or derived constructions. Thus topological vector spaces can serve as targets for ordinary symmetric-monoidal TQFTs while remaining poorly suited to these homological enhancements.

1.2. **The Condensed Fix.** Clausen and Scholze's condensed mathematics [Sch19] provides an abelian replacement for this homological setting. The category of condensed abelian groups (sheaves of abelian groups on the coherent site of compact Hausdorff spaces) is an abelian category with all limits and colimits. Every morphism has a kernel and cokernel. Every monic epic is an isomorphism.

Liquid vector spaces, a refinement of condensed modules, admit an exact completed tensor product. This was the content of the Liquid Tensor Experiment [Sch22, CT+22], Lean-verified by Commelin et al. in 2022. This exactness is valuable for homological gluing constructions, although it is stronger than the axioms of an ordinary TQFT and is not established by the present codebase.

1.3. **This Paper.** We report on a formal verification effort investigating whether CondensedAb can serve as a TQFT target category. The work proceeded through successive stages using Harmonic's Aristotle system [ABD+25], Claude (Anthropic), and OpenAI Codex for Lean 4 proof search, mathematical synthesis, API reconnaissance, and independent proof auditing.

The project produced both expected confirmations and genuine mathematical surprises. The monoidal structure on CondensedAb turned out to already exist in Mathlib. The embedding from seminormed groups turned out not to be full: a counterexample found while attempting to formalize fullness shows the obstruction, and we machine-checked it, first at the presheaf level and then for the sheaf-level functor itself. The later stages then ran an explicit Frobenius theory through the combinatorial semantics, closed the algebraic monoidal and symmetry gap in the source presentation, proved both reconstruction natural isomorphisms and the resulting universal equivalence, represented every finite boundary permutation by adjacent swaps, promoted the component/genus gluing model to a symmetric monoidal category, descended its signature as a strong braided functor, transported the concrete closed-word computations through the packaged symmetric-quotient theory, generalized the diagonal theory to arbitrary finite label types with relabeling isomorphisms, isolated a conditional ordinary-shadow reconstruction interface, constructed locally compatible tangent-orientation data and an honest boundary-parametrized cylinder, added a boundary homeomorphism and chosen-collar orientation substrate, constructed the compact topological pushout along a parametrized seam with no unintended identifications, proved that pushout Hausdorff by a closed-kernel quotient argument and second countable by compact-fiber basis descent, and completed a ribbon-category proof of quantum-dimension multiplicativity.

## 2. The Formalization

2.1. **Session 1: Abstract TQFT Framework.** The first session established the categorical scaffolding in the file `LiquidTQFT.lean`.

**Definition 2.1.** An *abstract TQFT* with source category $S$ and target category $C$ (both braided monoidal) is a braided monoidal functor $Z : S \to C$.

```
structure AbstractTQFT
    (S : Type*) [Category S] [MonoidalCategory S] [BraidedCategory S]
```

```
    (C : Type*) [Category C] [MonoidalCategory C] [BraidedCategory C]
       where
  Z : Functor S C
  monoidal : Z.Monoidal
  braided : Z.Braided
```

We package braided monoidal functors, a level of generality that includes the symmetric case and interfaces with non-symmetric braided categories used as algebraic input to three-dimensional constructions. Modular tensor categories underlying Reshetikhin–Turaev invariants [RT91] are a principal example of such input; the resulting oriented TQFT itself has the usual symmetric-monoidal bordism source. Lurie's cobordism hypothesis [Lur09] provides the fully extended framework in which these structures are most naturally understood.

**Theorem 2.2** (Transfer, placeholder-free)**.** *If $T$ is a TQFT valued in $C$ and $F : C \to D$ is a braided monoidal functor, then $T \circ F$ is a TQFT valued in $D$.*

```
def AbstractTQFT.transfer
    (T : AbstractTQFT S C)
    (F : Functor C D) [hF : F.Monoidal] [hFb : F.Braided] :
    AbstractTQFT S D where
  Z := F.comp T.Z
  monoidal := by letI := T.monoidal; infer_instance
  braided := by letI := T.monoidal; letI := T.braided; infer_instance
```

The proof is Lean's typeclass inference composing the monoidal and braided structures. Additional placeholder-free results from this session: CondensedAb is abelian, has kernels, cokernels, and finite (co)limits, and satisfies the balanced category property (mono + epi = iso).

The session also recorded structural observations for homological enhancements: in an abelian category, short exact sequences support decomposition methods one may combine with cutting and gluing. These statements take exactness as a hypothesis. The substantive analytic claim that the liquid tensor product is exact depends on the LTE and is not verified in this codebase; it is an additional strengthening, not part of the ordinary TQFT axioms.

2.2. **Session 2: Manual Monoidal Construction.** The second session (`CondensedMonoidal.lean`) attempted a direct construction of the monoidal structure on CondensedAb, defining the tensor product as sheafification of the pointwise tensor:

$$F \otimes_{\mathrm{cond}} G = \mathrm{sheafify}(F \otimes_{\mathrm{presheaf}} G).$$

This yielded a complete `MonoidalCategoryStruct` (placeholder-free) and 5/10 `MonoidalCategory` axioms. The remaining 5 reduced to two "sheafification comparison" isomorphisms:

$$\mathrm{sheafify}(P \otimes Q) \cong \mathrm{sheafify}(\mathrm{sheafify}(P)^{\mathrm{val}} \otimes Q).$$

This analysis was essential for the breakthrough in Session 3. As this manual construction was entirely superseded, it is not included in the final repository; we describe it here because the failed attempt is what motivated the search that found the localization approach.

2.3. **Session 3: The Localization Breakthrough.** The third session discovered that `Mathlib.CategoryTheory.Sites.Monoidal` [Theb] already contained the complete infrastructure.

**Theorem 2.3** (Placeholder-free)**.** CondensedAb *is a symmetric monoidal category.*

The construction assembles four existing Mathlib components:

(1) **Presheaf monoidal structure**: pointwise tensor on $\mathrm{CompHaus}^{\mathrm{op}} \to \mathrm{Ab}$.
(2) **Sheafification as localization**: `presheafToSheaf` is a localization functor for the class $J.W$ of local isomorphisms.

(3) $W$ **is monoidal**: $J.W$ is closed under tensoring, proved via the internal hom (monoidal closed structure), not stalks.
(4) **Localized monoidal category**: general construction from [Thea].

```
instance condensedAb_W_isMonoidal :
    ((coherentTopology CompHaus).W
      (A := ModuleCat (ULift Int))).IsMonoidal :=
  GrothendieckTopology.W.monoidal

instance condensedAb_monoidalCategory :
    MonoidalCategory CondensedAb :=
  Sheaf.monoidalCategory _ _
```

Six lines, zero proof placeholders. The internal hom argument (rather than stalks) powers the `W.IsMonoidal` proof, meaning the result holds for any closed braided target category. A direct stalk-based approach has since been formalized by Riou (Mathlib PR #35584), providing an alternative proof path.

2.4. **Session 4: The Banach Embedding.** The fourth session (`BanachEmbedding.lean`) audited Mathlib for connections between normed spaces and condensed mathematics. No packaged bridge was found in the pinned Mathlib v4.28.0 source.

**Definition 2.4.** For a seminormed abelian group $V$, the presheaf banachPresheaf $V$ sends $S \in$ CompHaus to $C(S, V)$, the abelian group of continuous maps, with precomposition as functorial action.

**Theorem 2.5** (Placeholder-free)**.** banachPresheaf $V$ *is a sheaf for the coherent topology on* CompHaus*.*

The proof transfers the sheaf condition from the Type-valued `yonedaPresheaf` through the forgetful functor. The main technical obstacle was universe arithmetic, bridged using `hasLimitsOfSizeShrink` and `preservesLimitsOfSize_of_univLE`.

**Theorem 2.6** (Placeholder-free)**.** *There exists a functor* SemiNormedGrp $\to$ CondensedAb*.*

2.5. **Session 5: Faithful but Not Full.** The fifth session established the central structural property of the embedding.

**Theorem 2.7** (Placeholder-free)**.** *The constructed embedding is faithful but* ***not*** *full. Both statements are machine-checked for the same sheaf-level functor.*

*Proof.* Faithfulness is the injectivity of postcomposition on bounded maps, verified by evaluation at the one-point space: constant functions separate the points of any seminormed group, so distinct bounded maps induce distinct condensed morphisms.

For non-fullness, let $V = \mathbb{N} \to_0 \mathbb{Z}$ (finitely supported integer sequences) with the supremum norm. Because every nonzero integer entry has absolute value at least 1, the norm satisfies $\|a\| \geq 1$ for $a \neq 0$, so $V$ carries the *discrete* topology; consequently every additive map out of $V$ is continuous. The summation map $f(a) = \sum_n a_n$ is therefore a continuous additive homomorphism, but it is not bounded: the indicators $w_N$ of $\{0, \dots, N\}$ satisfy $\|w_N\| = 1$ while $f(w_N) = N+1$. Postcomposition by $f$ defines a genuine morphism banachCondensed $V \to$ banachCondensed $\mathbb{Z}$ in CondensedAb (no sheafification argument is required, since both objects are already sheaves by construction). If the embedding were full, this morphism would come from a bounded map agreeing with $f$; evaluation at the one-point space forces pointwise agreement, contradicting unboundedness. $\square$

The formalization proceeds in two layers. `FullnessCounterexample.lean` verifies the analytic core and a self-contained presheaf-level functor $E$ : SemiNormedGrp $\to$ PSh(CompHaus, Ab) that is faithful and not full. `SheafFullnessCounterexample.lean` then transports the counterexample (after a universe lift, under which the discreteness and norm computations are unchanged) to the sheaf-level functor itself, yielding $\neg$ semiNormedGrpToCondensedAb.Full directly. The key observation is that the sheaf condition is never needed: the source and target of the offending morphism are already sheaves, so the morphism can be built between them outright.

The discreteness of $V$ is the engine: the condensed setting sees *all* additive maps as morphisms, while SemiNormedGrp admits only bounded ones, so the embedding cannot be full. The entire faithful-but-not-full development is placeholder-free and uses no custom axioms.

**Root cause.** Continuous additive maps between seminormed abelian groups need not be bounded. The standard proof that "continuous linear implies bounded" requires scalar multiplication by a dense subfield. This suggests restricting the source to normed spaces over a nontrivially normed field; fullness of that variant is not formalized here.

The session also constructed SemiNormedGrp.hasFiniteProducts (placeholder-free and not available in the pinned Mathlib version) and proved that the embedding preserves both finite products and equalizers, hence all finite limits, routing through the fact that the sheaf inclusion creates limits. The embedding is therefore left exact, machine-checked end to end.

2.6. **Session 6: Combinatorial Cobordisms.** The sixth session (`Cob2.lean`) formalized the algebraic structure classifying 2-dimensional TQFTs [Koc03].

**Definition 2.8** (Placeholder-free)**.** A *Frobenius object* in a monoidal category extends Mathlib's monoid and comonoid objects with the Frobenius relation:

$$(\mu \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \delta) = \delta \circ \mu.$$

A *commutative Frobenius object* additionally requires $\mu \circ \beta = \mu$.

**Theorem 2.9** (Placeholder-free)**.** *In any braided monoidal category, the unit object carries a canonical commutative Frobenius algebra structure.*

We define a combinatorial category $\mathrm{Cob}_2$ on the natural numbers, with morphisms generated by $\mu$ (pants), $\eta$ (cap), $\delta$ (copants), $\varepsilon$ (cup), plus composition, tensor, and swap, quotiented by the category axioms and the commutative Frobenius equations (with congruence closure). We emphasize the scope of this construction: tensor interchange, monoidal coherence on morphisms, and braiding naturality are *not* imposed, so this quotient is a preliminary presentation, coarser than a full presentation of the 2-dimensional cobordism category, and it carries no monoidal structure of its own.

**Theorem 2.10** (Placeholder-free)**.** *Every commutative Frobenius datum $A$ in a braided monoidal category $C$ induces a functor $\mathrm{Cob}_2 \to C$ sending $n$ to $X^{\otimes n}$ and the generators to the corresponding structure maps of $A$.*

This is a first algebraic rung toward one direction of the Frobenius classification of 2d TQFTs [Koc03], formalized via a recursive interpretation of generator words together with a proof that the interpretation respects every imposed relation. The base category `Cob2Cat` intentionally remains an ordinary preliminary quotient. Session 9 below strengthens its relation to obtain lawful monoidal and symmetric quotients, and Session 11 proves their algebraic commutative-Frobenius universal property. No identification with geometric oriented bordisms is asserted.

2.7. **Session 7: The Full Profile of the Embedding.** Session 7 (`EmbeddingProfile.lean`) added two machine-checked negative results and completed the embedding's structural profile.

**Theorem 2.11** (Placeholder-free)**.** *The embedding does not reflect isomorphisms. Explicitly, the $\ell^1$ and supremum norms on finitely supported integer sequences both induce the discrete topology, so the bounded identity between them realizes to an isomorphism in* CondensedAb *(the set-theoretic* *inverse is continuous and additive), yet it is not an isomorphism in* SemiNormedGrp*: any bounded inverse is refuted on the indicator vectors $w_N$ of $\ell^1$ norm $N+1$ and sup norm $1$.*

**Theorem 2.12** (Placeholder-free)**.** *The embedding does not preserve epimorphisms, and in particular is not right exact. Explicitly, equip $\mathbb{R}$ with the discrete group norm $\|x\|_d = |x| + [x \neq 0]$; the bounded surjective identity onto euclidean $\mathbb{R}$ is an epimorphism in* SemiNormedGrp*, but its realization fails the local-surjectivity criterion for epimorphisms of condensed modules: the section $[0,1] \hookrightarrow \mathbb{R}$ admits no local lift, since a continuous map from a compact space to a discrete space has finite image.*

Together with the earlier results, this yields a complete profile: the embedding is **additive, faithful, and left exact, but not full, not conservative, and not right exact**. The asymmetry (left exact but not right exact) is precisely the sense in which the condensed category re-topologizes the normed world: quotients that are invisible to bounded maps acquire genuine local structure on the condensed side.

2.8. **Session 8: A Concrete Diagonal Frobenius Theory.** The file `DijkgraafWitten.lean` supplies the first nontrivial input to the combinatorial semantics. For each $n$, let

$$A_n = \mathbb{Z}^{\mathrm{Fin}(n)}$$

with pointwise multiplication. On the standard basis $e_i$, the structure maps are

$$\eta(1) = \sum_i e_i, \qquad \delta(e_i) = e_i \otimes e_i, \qquad \varepsilon(x) = \sum_i x_i.$$

The construction uses `Pi.basisFun`, `Basis.constr`, and Mathlib's tensor product API. Associativity, the unit and counit laws, coassociativity, both Frobenius equations, and commutativity are proved by basis reduction.

**Theorem 2.13** (Diagonal torus and genus evaluation, placeholder-free)**.** *For the rank-$n$ datum `frobZn`, the closed word*

$$\varepsilon \circ \mu \circ \delta \circ \eta$$

*evaluates on $1 \in \mathbb{Z}$ to $n$. Equivalently, its image is multiplication by $n$ on the monoidal unit. More generally, every connected closed genus-$g$ presentation word defined in the file has the same value $n$.*

The generic calculation expresses a genus word as unit, a power of the handle operator $\mu \circ \delta$, and counit. For $A_n$ the handle operator is the identity, so all handle powers collapse and the remaining coordinate sum is $n$. This is a concrete finite-state Frobenius toy theory. It is not the conventional finite-group Dijkgraaf–Witten state-sum construction [DW90], and until the geometric comparison is established its theorem is an evaluation of algebraic presentation words rather than a diffeomorphism invariant of smooth surfaces. Session 10 transports these same scalar morphism equalities through the symmetric algebraic quotient without changing this geometric limitation.

2.9. **Session 9: Lawful Monoidal and Symmetric Cobordism Quotients.** The files `Cob2Monoidal.lean` and `Cob2Symmetric.lean` close the algebraic coherence gap identified in Session 6. The first enlarges the relation by tensor identities, interchange, and associator and unitor naturality. The resulting quotient is a lawful monoidal category; its pentagon and triangle equations follow from the arithmetic equality transports used for structural morphisms.

**Theorem 2.14** (Strong monoidal interpretation, placeholder-free)**.** *Every commutative Frobenius datum in a braided monoidal category induces a strong monoidal functor from the strengthened monoidal quotient, and precomposition with the canonical quotient functor recovers the original interpretation.*

The symmetric rung further imposes swap naturality, swap involutivity, and both braided hexagons.

**Theorem 2.15** (Symmetric algebraic source, placeholder-free)**.** *The strengthened quotient is a symmetric monoidal category. Every commutative Frobenius datum in a symmetric monoidal target induces a strong braided monoidal functor from this quotient. Passing through the symmetric quotient recovers the monoidal interpretation.*

Lean packages this result as `toSymmetricTQFT2d`. Session 10 instantiates this package with the diagonal datum and computes its transported closed presentation classes.

This completes the intended algebraic interchange-and-braiding rung. These two files alone do not establish the free-commutative-Frobenius-object theorem; Session 11 supplies that algebraic universal property. Equivalence with compact oriented smooth surfaces up to boundary-preserving diffeomorphism remains a separate, unproved geometric statement.

2.10. **Session 10: The Diagonal Theory on the Symmetric Quotient.** The new transport module composes the two existing ordinary quotient functors,

$$\begin{aligned}\mathrm{Cob2Cat} &\longrightarrow \mathrm{Cob2MonoidalObj}\\ &\longrightarrow \mathrm{Cob2SymmetricObj},\end{aligned}$$

to define `Cob2.toSymmetricQuotient`. For every commutative Frobenius datum $A$ in a symmetric target, interpretation after this composite is proved equal, as an ordinary functor, to `A.toCob2Functor`; on raw representatives the two maps are definitionally equal. No monoidal structure on this composite quotient functor is asserted.

For the diagonal datum, `frobZnSymmetricTQFT` packages `toSymmetricTQFT2d`. Its underlying functor is strong monoidal and braided, with symmetric source and target. The presentation classes `symmetricTorus` and `symmetricGenus g` are defined by transporting the already verified base classes, and `symmetricGenus 1 = symmetricTorus`.

**Theorem 2.16** (Symmetric-quotient diagonal evaluation, placeholder-free)**.** *For every $n, g \in \mathbb{N}$, if $Z_n$ denotes the underlying functor of `frobZnSymmetricTQFT n`, then*

$$Z_n(\texttt{symmetricTorus}) = (n : \mathbb{Z})\,\mathrm{id}_{\mathbf{1}}, \qquad Z_n(\texttt{symmetricGenus g}) = (n : \mathbb{Z})\,\mathrm{id}_{\mathbf{1}}.$$

*In particular, both underlying module morphisms evaluate at $1 \in \mathbb{Z}$ to $n$.*

These proofs transport the scalar morphism equalities from `DijkgraafWitten.lean`; they do not repeat the basis or tensor-product calculation. The result is a concrete strong monoidal, braided algebraic theory between symmetric categories. It does not identify the source with geometric oriented bordisms or establish classification, diffeomorphism, or gluing invariance. The `TQFT2d` package stores `Functor.Monoidal` and `Functor.Braided`; Lean has no additional field named `Functor.Symmetric` here.

The follow-up file `DijkgraafWittenDisconnected.lean` tensors a specified list of the connected genus words. If the list has $k$ entries, its ordinary and symmetric-quotient interpretations are both proved equal to $n^k\,\mathrm{id}_{\mathbf{1}}$. This is a calculation for those chosen presentation words, not a consequence of a classification of all disconnected surfaces.

2.11. **Session 11: Universal Equivalence, Boundary Permutations, and Surface Codes.** This algebraic checkpoint develops complementary views of the symmetric presentation. First, `Cob2Canonical.lean` equips the arity-one object with the four generating morphisms and proves that they form a commutative Frobenius datum internal to `Cob2SymmetricObj`. Its iterated tensor powers are canonically identified with the arity objects.

**Theorem 2.17** (Canonical-source reconstruction, placeholder-free)**.** *Interpreting the canonical arity-one Frobenius datum reconstructs the identity functor on `Cob2SymmetricObj`. The comparison is proved both as an ordinary natural isomorphism and as an isomorphism of bundled lax braided functors whose comparison maps are invertible.*

The proof establishes naturality by induction on every raw generator word, including tensor and swap, rather than appealing to geometric surfaces.

The files `Cob2Universal.lean` and `Cob2UniversalEquivalence.lean` make both sides of the algebraic universal property functorial and establish the first reconstruction isomorphism and the converse objectwise comparison.

`Cob2UniversalConverse.lean` proves the remaining naturality and completes the second reconstruction. Structure-preserving maps form a category of commutative Frobenius data. Strong braided functors are represented by the full subcategory of Mathlib's bundled lax braided functors for which the unit and tensor comparison maps are isomorphisms. Evaluation at the arity-one generator and interpretation of Frobenius data define functors

$$\mathrm{Interpret} : \mathrm{CommFrob}(C) \longrightarrow \mathrm{StrongBr}\big(\mathrm{Cob2SymmetricObj}, C\big),$$
$$\mathrm{Evaluate} : \mathrm{StrongBr}\big(\mathrm{Cob2SymmetricObj}, C\big) \longrightarrow \mathrm{CommFrob}(C).$$

**Theorem 2.18** (Algebraic commutative-Frobenius universal equivalence, placeholder-free)**.** *For every symmetric monoidal target $C$, there are natural isomorphisms*

$$\mathrm{Evaluate} \circ \mathrm{Interpret} \cong \mathrm{id}_{\mathrm{CommFrob}(C)}, \qquad \mathrm{Interpret} \circ \mathrm{Evaluate} \cong \mathrm{id}_{\mathrm{StrongBr}(\mathrm{Cob2SymmetricObj}, C)}.$$

*Consequently,*

$$\mathrm{StrongBr}\big(\mathrm{Cob2SymmetricObj}, C\big) \simeq \mathrm{CommFrob}(C).$$

The converse comparison uses the strong functor's tensor-power maps and proves naturality first on all raw generators and then on arbitrary source morphisms. This is the full universal property of the algebraic generators-and-relations source; it neither uses nor proves a classification of smooth surfaces.

Next, `Cob2Spider.lean` defines one fixed ordered connected word by merging all inputs, inserting $g$ copies of the handle operator, and splitting to all outputs. Frobenius sliding gives the following exact composition calculation.

**Theorem 2.19** (Ordered-spider composition, placeholder-free)**.** *For $b > 0$,*

$$\mathrm{spider}(b, c, h) \circ \mathrm{spider}(a, b, g) = \mathrm{spider}\big(a, c, g + (b - 1) + h\big).$$

Here $b - 1$ is the number of cycles added by gluing two connected pieces along $b$ common circles. Derived adjacent-boundary transposition identities first appear in `Cob2Permutation.lean` and `Cob2BoundaryPermutations.lean`.

Three follow-up modules extend absorption to every adjacent boundary position, package finite words of those swaps, and prove their incoming, outgoing, and two-sided absorption by spiders. A fourth module uses generation of the finite symmetric group by adjacent transpositions to obtain a representative for every $p \in \mathrm{Perm}(\mathrm{Fin}(n))$.

**Theorem 2.20** (Finite boundary-permutation absorption, placeholder-free)**.** *For every finite incoming or outgoing boundary permutation, a noncomputably chosen adjacent-swap word representing it is absorbed by every ordered connected spider on that boundary; the corresponding two-sided statement also holds.*

This is an existence theorem for representatives. It does not prove that two words representing the same permutation evaluate to the same Cob2 morphism, so the chosen representatives are not asserted to define a categorical group action. Nor does boundary invariance say that every connected presentation morphism is a spider.

The surface-code track starts in `Cob2NormalForm.lean` with a finite `SurfaceCode`: a component set with incident input and output circles and a genus label on each component. Codes are quotiented by component relabeling to form `SurfaceNF`. In `Cob2SurfaceComposition.lean`, composition uses the connected components of the finite bipartite multigraph whose vertices are old components and whose indexed edges are the glued middle circles. Keeping edge indices is essential because parallel middle circles contribute separately to genus. Relabeling invariance proves that this operation descends to `SurfaceNF`.

**Theorem 2.21** (Euler-safe code gluing, placeholder-free)**.** *For every component $q$ of a gluing graph,*

$$\#V_q \leq \#E_q + 1, \qquad g_q = g_q^{\mathrm{old}} + \big(\#E_q + 1 - \#V_q\big).$$

*Thus the natural-number subtraction defining the cycle contribution is not truncated. Moreover, gluing connected codes across $b > 0$ middle circles produces the connected code of genus $g+(b-1)+h$.*

**Theorem 2.22** (Surface-code category and disjoint union, placeholder-free)**.** *Graph gluing on `SurfaceNF` obeys both identity laws and associativity, so the arity-wrapped type `SurfaceNFObj` is a category. Disjoint union defines a bifunctor*

$$\mathrm{SurfaceNFObj} \times \mathrm{SurfaceNFObj} \longrightarrow \mathrm{SurfaceNFObj}$$

*that preserves identities and composition. Its transported associativity and empty-code unit equations are proved.*

`Cob2SurfaceWiring.lean` supplies the transport bridge from arity equalities to explicit boundary maps. A finite boundary equivalence is represented by a disjoint family of genus-zero cylinders. Gluing such a wiring to either side of a surface normal form is proved to reindex the corresponding external boundary without changing its component or genus data. In particular, the structural `eqToHom` maps on arity objects agree with these explicit wiring morphisms.

**Theorem 2.23** (Symmetric surface normal forms, placeholder-free)**.** *`Cob2SurfaceSymmetric.lean` packages disjoint union as a lawful monoidal structure on `SurfaceNFObj`. Its associator and unitors are the verified arity transports, its braiding is the block-swap wiring, and the naturality, involutivity, and both hexagon identities give a `SymmetricCategory` instance.*

`Cob2SurfaceSignature.lean` evaluates every raw Cob2 word as a surface code, proves all nine original commutative-Frobenius generator equations for that evaluation, and descends it to an ordinary functor

$$\mathrm{Cob2Cat} \longrightarrow \mathrm{SurfaceNFObj}.$$

`Cob2SurfaceSignatureSymmetric.lean` proves that the same evaluation respects every strengthened monoidal and symmetric relation. It therefore descends to

$$\mathrm{Cob2SymmetricObj} \longrightarrow \mathrm{SurfaceNFObj}$$

as a strong braided monoidal functor. In `Cob2ConnectedReification.lean`, the signature of the ordered spider with boundaries $(a, b)$ and genus $g$ is the canonical connected code with those data. Hence every canonical connected code is reified by an explicit word, including $a = 0$, $b = 0$, and $g = 0$, and the genus parameter of connected codes at fixed boundaries is injective. These results prove sound symmetric combinatorial semantics; they do not classify arbitrary words, prove uniqueness of their codes, establish that the signature is full or faithful, or give an equivalence with a geometric bordism category.

2.12. **Session 12: Geometric Boundary Data.** The file `Cob2GeometricPrelude.lean` begins the geometric rung with the smallest substrate supported honestly by the pinned Mathlib manifold API. It bundles compact smooth one-manifolds without model boundary and compact smooth two-manifolds modelled on a product with a half-space. A parametrized smooth cobordism from $M$ to $N$ is a single such surface together with a smooth embedded map from the disjoint union $M \sqcup N$ whose image is exactly the model-theoretic boundary.

**Theorem 2.24** (Cylinder boundary, placeholder-free). *For a closed smooth one-manifold $M$, the carrier $M \times [0,1]$ is a compact smooth surface with boundary, and its model boundary is*

$$\partial\big(M \times [0,1]\big) = M \times \{0,1\}.$$

`Cob2OrientedGeometricPrelude.lean` adds the first honest orientation-sensitive layer. A `SmoothTangentOrientation` chooses an orientation in every tangent space and requires the choices to be locally constant after transport through a tangent-bundle trivialization. Orientation reversal is involutive, and `PreservesDiffeomorph` states pointwise preservation through the differential of a smooth diffeomorphism. The file packages oriented closed one-manifold and compact surface carriers.

**Theorem 2.25** (Boundary-parametrized cylinder, placeholder-free). *For every stored closed smooth one-manifold $M$, the coproduct of endpoint inclusions*

$$M \sqcup M \longrightarrow M \times [0,1]$$

*is smooth, is a manifold immersion, is a topological embedding, and has image exactly $\partial(M \times [0,1])$. Consequently `cylinderCobordism M` is a verified `ParametrizedSmoothCobordism M M`.*

The immersion in `Cob2GeometricCylinder.lean` is proved directly in the canonical coproduct and product charts, with the interval tangent line as a global complement. This packages verified surface and boundary data; it does not prove that the cylinder is an identity for a composition law.

`Cob2GeometricBoundaryCollar.lean` then promotes the stored boundary parametrization of every `ParametrizedSmoothCobordism` to a homeomorphism onto the model-boundary subtype. It packages a chosen collar as a smooth topological embedding from $(M \sqcup N) \times [0,1]$ whose zero slice is the boundary parametrization and which is a local diffeomorphism along that slice.

**Theorem 2.26** (Collar image is a neighborhood). *For every supplied `BoundaryCollar` and every parametrized boundary point, the image of the collar belongs to the neighborhood filter of the corresponding boundary point. Thus the bundled data is a genuine collar neighborhood, not merely a smooth map meeting the boundary.*

The same file fixes a boundary-first, positive-inward-normal product orientation. Incoming one-manifold orientations are negated, outgoing orientations are retained, and `BoundaryCollar.IsOrientationCompatible` requires the collar differential to transport this product orientation to the stored surface orientation.

`Cob2GeometricTopologicalGluing.lean` takes the next composition step without assuming a smooth structure that has not been constructed. Given chosen-collared oriented cobordisms $W : M \to N$ and $V : N \to P$, their outgoing and incoming boundary maps form a span in `TopCat`. The file defines its categorical pushout as the topological gluing carrier.

**Theorem 2.27** (Compact topological seam pushout, placeholder-free). *The canonical maps from the carriers of $W$ and $V$ into the pushout are injective. A point $w \in W$ and a point $v \in V$ have the same pushout image if and only if there is a unique $x \in N$ whose outgoing and incoming boundary images are $w$ and $v$, respectively. The unglued $M \sqcup P$ outer-boundary map is injective, and the pushout carrier is compact.*

The file also exposes the pushout descent and uniqueness laws. After the two chosen half-collars are mapped into the pushout, their zero slices agree pointwise; the retained outgoing orientation of $W$ is the negative of the reversed incoming orientation of $V$ along the same seam. These are exact topological and orientation-level prerequisites for composition, not a claim that smooth composition already exists.

`Cob2GeometricHausdorffGluing.lean` proves the next separation property without adding any manifold or smooth structure. It exposes the canonical map

$$W \sqcup V \longrightarrow W \amalg_N V$$

from the topological sum of the two surface carriers and verifies directly that the topology on the categorical pushout is its quotient topology. The kernel relation is identified with the union of four compact closed subsets: the two diagonals and the two directed graphs of the common seam parametrization.

**Theorem 2.28** (Hausdorff topological seam pushout, placeholder-free)**.** *The quotient of a compact Hausdorff space by a surjective quotient map with closed kernel relation is Hausdorff. Applying this general theorem to the canonical sum quotient gives*

$$\mathtt{T2Space}\big(W \amalg_N V\big)$$

*for the topological gluing carrier of any two chosen-collared oriented parametrized smooth cobordisms with common boundary $N$.*

The general quotient theorem first proves that the quotient map is closed by expressing the saturation of a closed subset as the projection of a compact closed subset of the square. Normality of the compact Hausdorff source then separates two closed fibers, and closedness of the quotient map converts those separating sets into disjoint neighborhoods in the quotient.

`Cob2GeometricSecondCountableGluing.lean` then closes the global countability requirement. The argument is deliberately stated independently of the gluing construction. For a continuous surjection $f : X \to Y$ from a compact second-countable space to a Hausdorff space, compactness and Hausdorffness make $f$ closed. Starting from a countable basis of $X$, take kernel images under $f$ of finite unions of basis elements. These sets are open because $f$ is closed. Compactness of every fiber supplies a finite subcover, proving that this countable family is a basis of $Y$.

**Theorem 2.29** (Second-countable topological seam pushout, placeholder-free)**.** *Every continuous surjection from a compact second-countable space onto a Hausdorff space gives the target a second-countable topology. Applying this to the canonical sum quotient and the preceding Hausdorff theorem gives*

$$\mathtt{SecondCountableTopology}\big(W \amalg_N V\big)$$

*for the topological gluing carrier of any two chosen-collared oriented parametrized smooth cobordisms with common boundary $N$.*

The remaining boundary layer is not hidden. Mathlib's model boundary is still only a subset, so the development does not equip it with its own smooth manifold structure or prove that arbitrary surfaces admit the bundled collar. In particular it has not constructed an explicit collar for the cylinder: that requires chart-level smooth and local-diffeomorphism results for affine maps of the interval which the pinned API does not package. For the pushout, a local Euclidean-with-boundary structure, a compatible smooth seam atlas, smooth composition, identity and associativity up to boundary-preserving diffeomorphism, the cobordism category, and comparison with the algebraic source remain downstream.

2.13. **Session 13: Ribbon Categories and Quantum Dimension.** The root-level file `Ribbon.lean` develops an input-side abstraction relevant to three-dimensional TQFT. A balanced monoidal category is equipped with a natural twist satisfying the balancing equation, and a ribbon category additionally has chosen right duals with twist compatible with dualization. Every symmetric right-rigid monoidal category receives the identity-twist ribbon structure.

The file defines the quantum trace $\mathrm{qTr}(f)$, quantum dimension $\mathrm{qdim}(X) = \mathrm{qTr}(\mathrm{id}_X)$, and the $S$-pairing as the quantum trace of the double braiding. It proves trace cyclicity by sliding morphisms through the cup–braid–twist–cap diagram.

**Theorem 2.30** (Quantum-dimension multiplicativity, placeholder-free)**.** *In every ribbon category,*

$$\mathrm{qdim}(X \otimes Y) = \mathrm{qdim}(X) \circ \mathrm{qdim}(Y) \quad \textit{in } \mathrm{End}(\mathbf{1}).$$

*Moreover the $S$-pairing is symmetric.*

The multiplicativity proof constructs the reversed tensor-product exact pairing, verifies both triangle identities, factors the balanced evaluation using three Yang–Baxter rewrites, and moves the resulting scalar through the tensor factors. This is useful ribbon input infrastructure, but it does not construct a modular tensor category, prove nondegeneracy of the $S$-pairing, or define Reshetikhin–Turaev 3-manifold invariants.

2.14. **Session 14: Finite Label Symmetries and Conditional Ordinary Shadows.** `FiniteDiagonalFrobenius.lean` generalizes the rank-$n$ calculation without assigning additional geometry to its labels. For a commutative ring $R$ and a finite type $\iota$, it constructs the function algebra $R^\iota$ with pointwise multiplication, diagonal comultiplication on the delta-function basis, constant-function unit, and coordinate-sum counit.

**Theorem 2.31** (Finite-labelled diagonal theory, placeholder-free)**.** *The diagonal datum on $R^\iota$ is special: its handle operator is the identity. Every defined connected genus word therefore acts on the monoidal unit by the scalar $\#\iota$. Every permutation of $\iota$ induces an isomorphism of commutative Frobenius data, and interpretation sends it to a monoidal natural automorphism of the associated algebraic symmetric theory.*

This is a label-cardinality theorem with relabeling symmetry. It does not identify $\iota$ with a finite gauge group, construct a cocycle state sum, or turn the presentation-word calculation into a smooth-surface invariant.

`Cob2OrdinaryShadow.lean` isolates a separate conditional comparison interface. Its bridge structure consists of three supplied strong braided monoidal theories: a realization of the algebraic symmetric surface presentation, a selected intermediate sector, and a final realization or decategorification into a symmetric target. Their composite is again a `TQFT2d`; evaluation at the generating circle produces a commutative Frobenius datum.

**Theorem 2.32** (Conditional ordinary-shadow reconstruction, placeholder-free)**.** *For every supplied bridge, interpreting the generating-circle Frobenius datum is naturally isomorphic, as a strong braided functor, to the composite ordinary theory.*

The theorem is an application of the verified algebraic universal property once the three functors are provided. It constructs none of the geometric, Betti, de Rham, quantum, or decategorification functors that a Langlands or Fourier–Mukai application would require, and it asserts no such equivalence.

## 3. Results Summary

3.1. **Theorem Inventory.** Table 1 records the machine-checked statements used in this paper.

3.2. **Completeness of the Formalization.** The formalization contains no executable proof-admission placeholders and introduces no custom axioms: every formal theorem reported here is fully machine-checked. Dependency audits for the new headline theorems report only Lean's standard `propext`, `Classical.choice`, and `Quot.sound`.

Both limit-preservation results for the embedding were proved through a common architecture. `PreservesFiniteProducts` routes through the fact that the sheaf inclusion creates limits. `PreservesEqualizers` follows the same path: the canonical equalizer fork in SemiNormedGrp (vertex $\ker(f-g)$ with the subspace topology, already available in Mathlib's kernels file) is sent to a limit fork of types by each $C(S,-)$, since an equalizing continuous map lands in the kernel and corestricts continuously; the statement then lifts pointwise through evaluation and through the sheaf inclusion. Consequently the embedding preserves all finite limits: **the embedding** SemiNormedGrp $\to$ CondensedAb **is left exact, machine-checked end to end.**

The base functor toCob2Functor was completed by recursively interpreting raw generator words as morphisms between tensor powers. The tensor case uses comparison isomorphisms $X^{\otimes(a+c)} \cong$

$X^{\otimes a} \otimes X^{\otimes c}$, built by induction from associators and the right unitor; the monoid and comonoid generators are conjugated by unitors to match this convention. Soundness over the original presentation equations is proved constructor by constructor, after which the functor descends via `Quotient.lift`.

The two strengthened quotients then close the algebraic coherence gap. `Cob2MonoidalRel` adds tensor identity, interchange, and structural naturality; coherence of the same `powAdd` comparisons supplies a strong monoidal interpretation. `Cob2SymmetricRel` adds swap naturality, involutivity, and both hexagons; interpretation into a symmetric target is sound and braided. The compatibility theorems show that passing from the base quotient through the monoidal and symmetric rungs recovers the earlier semantics. The composite `Cob2.toSymmetricQuotient` packages both steps, and interpretation after it is proved equal to the base semantics for every commutative Frobenius datum.

The canonical arity-one datum supplies the internal reverse check: its interpretation is isomorphic to the identity of the symmetric algebraic source, including in the braided monoidal bundle. Ordered connected spiders then satisfy the expected positive-boundary composition law. Every finite boundary permutation has an adjacent-swap word representative, and a chosen representative is absorbed at either boundary of a spider. The result is intentionally existential: evaluation is not proved independent of the representing word, and no group action is claimed.

At the categorical level, evaluation at the generator and interpretation are both functors. Both composites are naturally isomorphic to the corresponding identity functor, yielding an equivalence between strong braided functors from `Cob2SymmetricObj` and commutative Frobenius data in every symmetric target. This equivalence is purely algebraic and does not depend on surface classification.

Independently, component/genus codes form a quotient by relabeling on which finite-multigraph gluing is well-defined; a connected-simple-graph Euler estimate verifies that the cycle-rank subtraction in the genus is nontruncated. Both unit laws and associativity promote the arity-wrapped normal forms to a category. Disjoint union is a bifunctor satisfying interchange, identity, transported associativity, and empty-code unit equations. Genus-zero boundary wirings identify those arity transports with explicit surface codes, allowing the bifunctor to be packaged as a lawful symmetric monoidal structure whose braiding is block swap. Evaluation of raw Cob2 words as codes respects not only the original Frobenius relation but also every strengthened monoidal and symmetric relation, and hence descends as a strong braided functor from the symmetric algebraic quotient. Canonical connected codes are reified by ordered spiders and have injective genus, but arbitrary-word completeness, uniqueness, and any equivalence theorem remain open.

The geometric layer is separate from the algebraic presentation. It packages closed smooth one-manifolds, compact smooth surfaces with boundary, and individual boundary-parametrized cobordisms; defines tangent orientations by local compatibility in tangent-bundle trivializations; proves that the endpoint coproduct embeds smoothly onto the cylinder boundary; identifies every stored parametrized boundary homeomorphically with the model-boundary subtype; and packages chosen collars with a proved neighborhood property and an incoming-reversed/outgoing-retained orientation-compatibility predicate. For two chosen-collared pieces it then constructs the compact topological pushout, proves both piece maps and the outer boundary injective, characterizes cross-piece equality by a unique seam point, and verifies the universal and collar/orientation seam laws. The canonical map from the sum of the two carriers is proved to be a quotient map; its kernel is the union of the two diagonal and two directed seam-graph relations, all compact and closed. A general closed-kernel quotient theorem for compact Hausdorff sources then proves that the gluing carrier is Hausdorff. A second general theorem descends a countable basis along a continuous surjection from a compact second-countable space to a Hausdorff target, and its application proves that the gluing carrier is second countable. The development does not make the boundary or pushout a smooth manifold in its own right, prove collar existence, construct an explicit cylinder collar, establish a local Euclidean-with-boundary structure, provide a compatible smooth atlas or smooth gluing, prove the cylinder is an identity, define composition or diffeomorphism classes, form a cobordism category, or establish any equivalence with the algebraic source.

The diagonal basis calculation is performed once in `DijkgraafWitten.lean`. `DijkgraafWitten Symmetric.lean` reuses it to prove the scalar morphism equalities through the underlying functor of the packaged symmetric-quotient theory; no basis-level computation is duplicated. `FiniteDiagonalFrobenius.lean` repeats the construction at its proper generality—a commutative coefficient ring and an arbitrary finite label type—and proves that genus words see only label cardinality while label permutations act by Frobenius isomorphisms. `Cob2OrdinaryShadow.lean` then records a conditional application of the universal property to any supplied chain of strong braided theories; it does not construct or prove equivalence of the intended geometric or quantum inputs. The ribbon proof builds and verifies the reversed tensor pairing explicitly; a six-crossing braid equality, resolved by three Yang–Baxter moves, is the key step in factoring the quantum cap and proving multiplicativity.

3.3. **Key Findings.** **Finding 1.** The monoidal structure on CondensedAb was already in Mathlib. The iterative process (axiom $\rightarrow$ manual construction $\rightarrow$ localization discovery) demonstrates the difficulty of navigating large formal libraries.

**Finding 2.** No packaged bridge was found between Mathlib's `Analysis.NormedSpace` and `Condensed` components in v4.28.0. The project supplies a concrete connection.

**Finding 3.** Fullness fails for the embedding (Theorem 2.7), machine-checked for the sheaf-level functor itself. The condensed setting sees strictly more morphisms than the normed category, because the sup-norm topology on integer sequences is discrete and admits unbounded continuous additive maps.

**Finding 4.** SemiNormedGrp.hasFiniteProducts was not available in the pinned Mathlib v4.28.0 source.

**Finding 5.** The Frobenius-generator semantics extends to a lawful symmetric monoidal algebraic source and a strong braided monoidal interpretation. The canonical composite from the base quotient to the symmetric quotient is semantically compatible with the original interpretation for every commutative Frobenius datum. The canonical arity-one datum gives both natural reconstruction isomorphisms, and strong braided functors from the symmetric algebraic source are categorically equivalent to commutative Frobenius data in every symmetric target. This completes the algebraic universal property while leaving geometric identification open.

**Finding 6.** The rank-$n$ diagonal Frobenius theory is executable mathematics: the torus and every connected genus word in the defined family act by multiplication by $n$, and a specified list of $k$ such components acts by $n^k$. These scalar morphism equalities persist through the underlying functor of the packaged strong monoidal, braided theory on the symmetric algebraic quotient. Over any commutative ring, the same diagonal construction on a finite label type evaluates connected genus words to the label cardinality, and every label permutation acts by a Frobenius isomorphism.

**Finding 7.** Ribbon-category string-diagram identities can be proved directly against Mathlib's associator, braiding, and exact-pairing APIs. In particular, quantum dimension is multiplicative and the $S$-pairing is symmetric without assuming modularity.

**Finding 8.** The genus correction in algebraic surface-code composition is the cycle rank of a finite gluing multigraph. Retaining indexed parallel edges and applying an Euler bound to the underlying connected simple graph proves that the natural-number formula does not truncate. Explicit genus-zero wirings turn disjoint union into a lawful symmetric monoidal structure, and the surface signature descends from the symmetric algebraic source as a strong braided functor. Canonical connected codes remain reified by ordered spiders. This is sound semantics, not yet a complete or unique classification of arbitrary presentation words or an equivalence of categories.

**Finding 9.** Mathlib's current manifold infrastructure supports an honest substrate for individual two-dimensional cobordisms, locally compatible tangent orientations, and a smoothly embedded endpoint parametrization of the cylinder. The cylinder is therefore packaged as a boundary-parametrized smooth cobordism. The boundary parametrization is further promoted to a homeomorphism onto the boundary subtype; chosen smooth collars are packaged with a proved neighborhood property, and their differentials can be required to realize reversed incoming and retained outgoing orientations. Two chosen-collared pieces admit a verified compact Hausdorff second-countable topological pushout with injective piece and outer-boundary maps, a unique-seam characterization of cross-piece identifications, the pushout universal property, and compatible zero-slice and boundary-orientation equations. Hausdorffness follows by identifying the canonical sum quotient's kernel with four compact closed relations and applying a general compact-Hausdorff closed-kernel quotient theorem. Second countability follows by descending finite unions of a countable source basis along the closed canonical quotient map, using compactness of each fiber. A local Euclidean-with-boundary structure and compatible smooth atlas, collar existence, an explicit cylinder collar, smooth composition, identity and associativity laws, and the quotient-category construction remain distinct missing layers.

**Finding 10.** The algebraic universal property provides a precise conditional ordinary-shadow principle: once suitable strong braided comparison, sector, and realization functors are supplied, the composite theory is reconstructed from its generating-circle Frobenius datum. This separates the proved formal consequence from the unconstructed geometric and quantum inputs and does not itself establish a Langlands or Fourier–Mukai correspondence.

## 4. Roadmap for Further Development

4.1. **Near-term.** Prove that equivalent adjacent-swap words evaluate to the same boundary morphism and, if useful, package the resulting finite-permutation action. Determine the precise arbitrary-word completeness and uniqueness theorem for the strong braided surface signature, and separately investigate its fullness, faithfulness, and essential surjectivity without assuming any of them. Instantiate the finite-labelled diagonal module inside CondensedAb through the discrete embedding. For any proposed ordinary-shadow application, construct the comparison, sector, and realization functors required by `OrdinaryShadowBridge` and verify their strong braided structures rather than treating them as implicit.

4.2. **Medium-term.** Use the strong braided signature to compare the symmetric algebraic source with the component/genus normal-form category, proving only the categorical properties that can be derived from explicit normal-form theorems. On the geometric side, first prove the chart-level smooth and local-diffeomorphism lemmas for affine interval maps needed to construct an explicit two-ended collar of the cylinder. Starting from the verified compact Hausdorff second-countable seam pushout, next prove a local Euclidean-with-boundary structure, then build compatible signed collar charts across the seam. Investigate collar existence beyond the bundled-data interface only after keeping the distinction between supplied and constructed collars explicit. On the analytic side, define an appropriate category of normed spaces over a valued field and determine which fullness or exactness statements survive after imposing the hypotheses of the open mapping theorem.

4.3. **Long-term.** Promote the topological pushout and seam atlas to smooth gluing from chosen compatible collars, prove its unit and associativity laws up to boundary-preserving orientation-preserving diffeomorphism, and form a geometric 2Cob category. Only then compare it with the algebraic source. Develop modular tensor category input and surgery/Kirby-move infrastructure for Reshetikhin–Turaev theory; `Ribbon.lean` supplies only the first input-side layer. Formalize the projective tensor product of normed spaces and connect to liquid tensor exactness.

4.4. **Frontier.** Construct mathematically rigorous non-compact Chern–Simons state spaces and functorial gluing before attempting their formalization in liquid or nuclear targets. A parallel frontier is Costello–Gwilliam [CG17] factorization algebras in the condensed setting.

## 5. Methodology

The work proceeded through successive rounds of AI-assisted formal verification. A typical round consisted of: (1) scoping a concrete mathematical target; (2) reconnaissance of the exact Mathlib API and categorical conventions; (3) construction and proof search; and (4) independent compilation, placeholder scanning, and axiom-dependency auditing. Aristotle and Claude supported earlier rounds; OpenAI Codex completed and independently audited the ribbon multiplicativity proof, the symmetric-quotient transport, the algebraic reconstruction, finite-permutation and surface-code checkpoint, geometric API audit, and repository integration. Aristotle supplied the final converse-reconstruction source, which was independently compiled and audited before integration.

The prompts evolved from exploratory to targeted. The most productive prompts front-loaded reconnaissance (searching Mathlib for existing infrastructure) before attempting construction. The Session 3 breakthrough directly resulted from this "search before build" approach.

### 5.1. Tools.

- **Lean 4** with **Mathlib v4.28.0** [The24] for formal verification.
- **Harmonic Aristotle** [ABD+25] for proof search and Mathlib API discovery.
- **Claude (Anthropic)** for mathematical synthesis, prompt engineering, and documentation.
- **OpenAI Codex** for proof completion, independent auditing, and repository integration.
- All formal results independently verifiable via `lake build`; headline proofs were additionally checked with `#print axioms`.

## 6. Conclusion

Successive rounds of AI-assisted formal verification, starting from a blank slate, produced 16,153 lines of proof-placeholder-free Lean 4 across 44 source files, with 0 custom axioms. The central results are machine-checked end to end: CondensedAb is a symmetric monoidal abelian category; the TQFT transfer theorem holds; the seminormed-group realization is faithful and left exact but not full, not conservative, and not right exact; the Frobenius presentation admits lawful monoidal and symmetric algebraic quotients with strong monoidal semantics; strong braided functors from the symmetric algebraic source are equivalent to commutative Frobenius data; ordered spiders obey the positive-boundary genus-composition law and absorb chosen representatives of every finite boundary permutation; component/genus codes form a lawful symmetric monoidal category with descended multigraph gluing, a verified nontruncated Euler contribution, a strong braided signature from the symmetric presentation, and connected reification; the rank-$n$ diagonal theory sends its transported symmetric torus and defined connected genus classes to $n$ times the identity, while its finite-labelled extension computes label cardinality and admits relabeling isomorphisms; any supplied ordinary-shadow bridge satisfies the universal reconstruction comparison; the geometric layer defines locally compatible tangent orientations, packages the smoothly embedded endpoint parametrization of the cylinder, identifies parametrized boundaries topologically, supplies chosen-collar neighborhood and orientation-compatibility data, constructs a compact topological seam pushout with exact identification and universal laws, proves it Hausdorff through a closed-kernel quotient theorem, and proves it second countable by compact-fiber basis descent; and quantum dimension is multiplicative in every ribbon category.

A notable result was machine-checking that the embedding SemiNormedGrp $\to$ CondensedAb is faithful but not full, with both directions verified for the sheaf-level functor itself. This is a precise mathematical boundary: the condensed setting sees more morphisms between normed groups than the normed category itself does, because the sup-norm topology on integer sequences is discrete and admits unbounded continuous additive maps. Normed vector spaces over a valued field provide the natural next setting because continuity of linear maps then controls boundedness; that variant is not formalized here.

The remaining barriers are now sharply separated by scale. The next algebraic steps are representation independence for finite permutation words, proving the correct arbitrary-word completeness and uniqueness theorem, and determining whether the strong braided surface signature has any stronger categorical properties. The ordinary-shadow interface still requires concrete comparison, sector, and realization functors before it says anything about a geometric or quantum duality. The next geometric steps are an explicit cylinder collar and, for the existing Hausdorff second-countable seam pushout, a local Euclidean-with-boundary structure and compatible signed collar charts; only after a compatible smooth atlas, smooth composition, and its coherence laws are proved may one claim diffeomorphism classes, a geometric cobordism category, and comparison with the algebraic quotient. The current geometric layer classifies no smooth surfaces and proves no diffeomorphism or smooth-gluing invariance, and the verified cylinder is not yet a categorical identity. Three-dimensional Reshetikhin–Turaev theory additionally requires modular tensor categories, surgery presentations, and Kirby moves. The motivating non-compact Chern–Simons target lies beyond these: its infinite-dimensional state spaces and gluing theory must first be constructed rigorously on paper before the liquid or nuclear machinery can be applied.

**Code Availability.** The paper and a synchronized Lean 4 snapshot are available at `https://github.com/seanm27lol/Liquid-TQFT-lean-CMCM`. The canonical implementation repository is `https://github.com/seanm27lol/Liquid-TQFT-CMCM-cont.`. The repository revisions, rather than an embedded stale commit hash, are authoritative for the checkpoint reported here. Both build against Mathlib v4.28.0.

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| **Result** | **File** | **Status** |
|---|---|---|
| CondensedAb is abelian | LiquidTQFT.lean | Placeholder-free |
| Mono + epi = iso | LiquidTQFT.lean | Placeholder-free |
| MonoidalCategory CondensedAb | LiquidTQFT.lean | Placeholder-free |
| BraidedCategory CondensedAb | LiquidTQFT.lean | Placeholder-free |
| SymmetricCategory CondensedAb | LiquidTQFT.lean | Placeholder-free |
| `AbstractTQFT.transfer` | LiquidTQFT.lean | Placeholder-free |
| $W$.IsMonoidal for coherent topology | MonoidalViaLoc.lean | Placeholder-free |
| `presheafToSheaf` is monoidal | MonoidalViaLoc.lean | Placeholder-free |
| `banachPresheaf` $V$ is a sheaf | BanachEmbedding.lean | Placeholder-free |
| SemiNormedGrp $\to$ CondensedAb functor | BanachEmbedding.lean | Placeholder-free |
| Embedding faithful (sheaf level) | BanachEmbedding.lean | Placeholder-free |
| Embedding not full (sheaf level) | SheafFullnessCounterexample.lean | Placeholder-free |
| Presheaf functor $E$ faithful, not full | FullnessCounterexample.lean | Placeholder-free |
| Embedding preserves finite products | BanachEmbedding.lean | Placeholder-free |
| Embedding preserves equalizers | BanachEmbedding.lean | Placeholder-free |
| Embedding is left exact | BanachEmbedding.lean | Placeholder-free |
| Embedding does not reflect isos | EmbeddingProfile.lean | Placeholder-free |
| Embedding does not preserve epis | EmbeddingProfile.lean | Placeholder-free |
| SemiNormedGrp.hasFiniteProducts | BanachEmbedding.lean | Placeholder-free |
| `FrobeniusObj`, `CommFrobeniusObj` | Cob2.lean | Placeholder-free |
| Trivial Frobenius on $\mathbf{1}_C$ | Cob2.lean | Placeholder-free |
| Frobenius datum $\Rightarrow$ functor from $\mathrm{Cob}_2$ | Cob2.lean | Placeholder-free |
| $\mathrm{Cob}_2$ category | Cob2.lean | Placeholder-free |
| Rank-$n$ diagonal Frobenius datum | DijkgraafWitten.lean | Placeholder-free |
| Torus and genus-word evaluations equal $n$ | DijkgraafWitten.lean | Placeholder-free |
| Finite-labelled diagonal datum and genus value $\#\iota$ | FiniteDiagonalFrobenius.lean | Placeholder-free |
| Label-relabeling Frobenius and interpreted isomorphisms | FiniteDiagonalFrobenius.lean | Placeholder-free |
| Lawful monoidal presentation quotient | Cob2Monoidal.lean | Placeholder-free |
| Strong monoidal Frobenius interpretation | Cob2Monoidal.lean | Placeholder-free |
| Symmetric monoidal presentation quotient | Cob2Symmetric.lean | Placeholder-free |
| Strong braided monoidal interpretation | Cob2Symmetric.lean | Placeholder-free |
| Base-to-symmetric interpretation compatibility | DijkgraafWittenSymmetric.lean | Placeholder-free |
| Packaged symmetric torus/genus maps equal $n$ id | DijkgraafWittenSymmetric.lean | Placeholder-free |
| Specified $k$-component words map to $n^k$ id | DijkgraafWittenDisconnected.lean | Placeholder-free |
| Canonical Frobenius and source reconstruction | Cob2Canonical.lean | Placeholder-free |
| Ordered positive-boundary spider composition | Cob2Spider.lean | Placeholder-free |
| Every finite boundary permutation is representable | Cob2FinitePermutationWords.lean | Placeholder-free |
| Represented boundary permutations are absorbed by spiders | Cob2SpiderPermutationInvariance.lean | Placeholder-free |
| Surface-code composition and Euler bound | Cob2SurfaceComposition/GraphBound | Placeholder-free |
| Surface-code category laws | Cob2SurfaceCategory.lean | Placeholder-free |
| Surface disjoint-union bifunctor and coherence | Cob2SurfaceMonoidal/Coherence | Placeholder-free |
| Boundary wiring and external reindexing laws | Cob2SurfaceWiring.lean | Placeholder-free |
| Lawful symmetric monoidal surface normal forms | Cob2SurfaceSymmetric.lean | Placeholder-free |
| Ordinary Cob2 surface signature functor | Cob2SurfaceSignature.lean | Placeholder-free |
| Strong braided symmetric-quotient surface signature | Cob2SurfaceSignatureSymmetric.lean | Placeholder-free |
| Connected-code reification and genus injectivity | Cob2ConnectedReification.lean | Placeholder-free |
| Full commutative-Frobenius universal equivalence | Cob2UniversalConverse.lean | Placeholder-free |
| Conditional ordinary-shadow reconstruction | Cob2OrdinaryShadow.lean | Placeholder-free |
| Smooth-cobordism carrier data and cylinder boundary | Cob2GeometricPrelude.lean | Placeholder-free |
| Locally compatible tangent orientations | Cob2OrientedGeometricPrelude.lean | Placeholder-free |
| Smooth embedded cylinder parametrization | Cob2GeometricCylinder.lean | Placeholder-free |
| Boundary homeomorphism, collar neighborhood, and oriented compatibility | Cob2GeometricBoundaryCollar.lean | Placeholder-free |
| Compact topological seam pushout with exact identifications | Cob2GeometricTopologicalGluing.lean | Placeholder-free |
| Hausdorffness of the compact topological seam pushout | Cob2GeometricHausdorffGluing.lean | Placeholder-free |
| Second countability of the compact topological seam pushout | Cob2GeometricSecondCountable Gluing.lean | Placeholder-free |
| Quantum-trace cyclicity and $S$-symmetry | Ribbon.lean | Placeholder-free |
| Quantum-dimension multiplicativity | Ribbon.lean | Placeholder-free |

TABLE 1. Theorem inventory for the machine-checked development. The faithful-but-not-full result (Theorem 2.7) is verified for the sheaf-level embedding itself. The cobordism quotients are algebraic presentations; geometric identification remains future work.